



The World Bank

Second Low-Carbon Energy Programmatic Development Policy Operation (P181195)

Document of
The World Bank

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Report No: PGD465

INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT

PROGRAM DOCUMENT FOR A

PROPOSED DEVELOPMENT POLICY LOAN

IN THE AMOUNT OF US\$1,500 MILLION TO

THE REPUBLIC OF INDIA

FOR THE

Second Low-Carbon Energy Programmatic Development Policy Operation

September 21, 2023

Macroeconomics, Trade And Investment Global Practice
Energy & Extractives Global Practice
South Asia Region

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Republic of India

GOVERNMENT FISCAL YEAR

April 1 – March 31

CURRENCY EQUIVALENTS

(Exchange Rate Effective as of September 15, 20223)

Currency Unit

US\$1.00 = INR 83.05

ABBREVIATIONS AND ACRONYMS

BESS	Battery Energy Storage System	MNRE	Ministry of New and Renewable Energy
CEA	Central Electricity Authority	MoF	Ministry of Finance
CCTS	Carbon Credit Trading Scheme	MoP	Ministry of Power
DPO	Development Policy Operation	NGHM	National Green Hydrogen Mission
EE	Energy Efficiency	PA	Prior Action
ESG	Environment, Social and Governance	PAT	Perform,-Achieve-Trade
FY	Fiscal Year	PDO	Project Development Objective
GDP	Gross Domestic Product	PLI	Production-Linked Incentive
GH	Green Hydrogen	PV	Photovoltaic
GHG	Greenhouse Gas	RBI	Reserve Bank of India
GW	Gigawatt	RE	Renewable Energy
GWh	Gigawatt-hour	RPO	Renewable Purchase Obligation
IMF	International Monetary Fund	SEBI	Securities and Exchange Board of India
INR	Indian Rupee	WB	World Bank
IBRD	International Bank for Reconstruction and Development		
ISTS	Inter-State Transmission System		

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INDIA

SECOND LOW-CARBON ENERGY PROGRAMMATIC DEVELOPMENT POLICY OPERATION

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SUMMARY OF PROPOSED FINANCING AND PROGRAM

BASIC INFORMATION

Project ID	Programmatic	If programmatic, position in series
P181195	Yes	2nd in a series of 2

Proposed Development Objective(s)

To accelerate the development of low-carbon energy in India.

Organizations

Borrower:	INDIA
Implementing Agency:	Ministry of New and Renewable Energy

PROJECT FINANCING DATA (US\$, Millions)

SUMMARY

Total Financing	1,500.00
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DETAILS

International Bank for Reconstruction and Development (IBRD)	1,500.00
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INSTITUTIONAL DATA

Climate Change and Disaster Screening

This operation has been screened for short and long-term climate change and disaster risks

Overall Risk Rating

Substantial



Results

Indicator Name	Baseline (June 30, 2023)	Current	Target (June 30, 2026)
Pillar 1: Promoting green hydrogen (GH)			
GH production capacity incentivized (tonnes)	0	0	450,000
Domestic electrolyzer manufacturing capacity incentivized (gigawatt, MW)	0	0	500
Number of additional GH safety standards notified	0	0	10
Pillar 2: Scaling up renewable energy (RE)			
Share of RE in power consumption (percentage)	25	25	33
Share of solar/wind with/through energy storage in total power consumption (percentage)	0	0	2
BESS incentivized	0	0	TBA
Additional bids for RE capacity issued (GW)	0	0	75
Greenhouse gas (GHG) emissions avoided (million tonnes per annum)	0	0	40
Offshore wind sites awarded (equiv. to power capacity GW)	0	0	4
Capacity for domestic manufacturing of high-efficiency solar photovoltaic cells and modules added (GW)	0	0	48
Pillar 3: Enhancing climate finance for low-carbon energy investment			
Launch of a national carbon market	No carbon market exists	Carbon Credit Trading Scheme issued	GHG emission intensity targets allocated and carbon trading operational
Cumulative additional issuance of onshore green debt securities (corresponding to 30% increase from 45.4 bn baseline)	0	0	13.6 billion
Sovereign green bonds issued (cumulative, US\$ billion)	2.3	2.3	6.0



IBRD PROGRAM DOCUMENT FOR A PROPOSED LOAN TO THE REPUBLIC OF INDIA

1. INTRODUCTION AND COUNTRY CONTEXT

1. **India has made very significant development progress in recent decades; fast demographic expansion and high aspirations mean it will need to continue to sustain high rates of growth going forward.** From 1991 to today, the size of the economy was multiplied over 10 times, and Gross Domestic Product (GDP) per capita by 7 times. Over 1993-2022, the (extreme) poverty rate declined by 36 percentage points—from 47.6 percent to 11.3 percent—reducing the number of poor people in India by over 280 million. India’s past achievements set the tone for its future ambitions, notably the target to reach high-income by 2047. However, getting there will require the economy to grow at 8 percent per annum over the next two decades.

2. **Growth objectives will need to be reconciled with equally ambitious climate commitments.** India has set itself the goal of achieving net-zero emissions by 2070 and adopted ambitious short-term targets by 2030¹. To simultaneously achieve its decarbonization and economic growth objectives, it will need to meet its rapidly growing energy needs while reducing emissions. Specifically, this will require: (a) electrifying end-user sectors (particularly transport, buildings and targeted industries); (b) greening the power sector with renewable energy (RE) and integrating RE in the grid; and (c) bringing hard-to-abate industries onto a low-carbon path, with GH and carbon capture.

3. **The GoI’s low-carbon energy strategy is to accelerate RE development to meet fast-growing energy demand, while eventually allowing the “phase-down” of fossil fuels.** The GoI has implemented a series of energy efficiency and RE policies, which have led to a paradigm shift toward RE in the power sector. Going forward, RE grid integration will be critical to accommodate a high share of RE in the power mix.

4. **Green Hydrogen (GH) can play a major role in India’s low carbon development pathway.** Beyond the power sector, industry will be the main driver of future growth in energy demand and GHG emissions. This is because high economic growth will go hand-in-hand with increased output from heavy industries (such as cement and steel) that -under the current trends- would gradually account for the majority of GHG emissions. GH will be critical to decarbonize the hard-to-abate industrial sectors, particularly the fertilizer, refinery, and steel sectors. However, the technology is still nascent. GH has not yet reached commercial viability and several uncertainties remain, including: technological challenges, high cost of production, transport and storage, and safety, environmental and social concerns. To overcome these obstacles and position India at the forefront of the GH development frontier, the government launched the National Green Hydrogen Mission (NGHM) in January 2023, with a total budget of US\$2.4 billion. The NGHM’s goals include helping to bridge the viability cost gap, and promoting public and private investments in GH.

¹ including: (a) reducing the share of its power capacity from non-fossil fuels to below 50 percent; (b) bringing the emissions intensity of GDP down by 45 percent from 2005 to 2030; and (c) creating additional cumulative carbon sinks of 2.5 to 3 billion tons of CO₂ through additional forest and tree covers (<https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1847812>)



2. MACROECONOMIC POLICY FRAMEWORK

2.1. RECENT ECONOMIC DEVELOPMENTS

5. **Growth decelerated to 7.2 percent in FY22/23, from 9.1 percent in FY21/22, mainly due to waning base effects.** Private consumption growth moderated as post-pandemic catch-up spending tapered off, and public consumption stagnated due to consolidation of recurrent spending. Investment growth remained robust on the back of the government's capital expenditure push, with much of this spending directed towards building infrastructure. Economic growth remained robust in the first quarter of FY23/24, at 7.8 percent.

6. **After declining steadily in FY22/23, headline inflation has risen recently on the back of a temporary increase in food prices.** It averaged 6.7 percent y-o-y in FY22/23 and trended down until May 2023, as commodity prices eased and the RBI hiked rates six times by 250 basis points cumulatively between May 2022 and February 2023. Although abnormal monsoon rainfall led to a large increase in food prices and headline inflation increased to 7.4 percent in July 2023, the RBI kept the policy rate unchanged on account of the transient nature of inflationary pressures.

7. **India's financial sector is in much better health than it was pre-COVID.** The gross non-performing assets ratio for banks has fallen to a decadal low of 3.9 percent² in March 2023, compared with over 11 percent in March 2018. Banks are adequately capitalized, with the capital-to-risk-weighted-asset ratio reaching a historical high of 17.1 percent, allowing banks to withstand severe macroeconomic shocks. The improvement in the balance sheets of banks and the corporate sector has enabled an acceleration in credit growth.

8. **The current account deficit (CAD) reached 2.0 percent of GDP in FY22/23 but narrowed in the first quarter of FY23/24.** Between April and June 2023, the CAD narrowed to 1.1 percent of GDP. Easing global commodity prices and softening demand in key export markets caused a simultaneous decline in goods imports and exports. However, India's services exports continued to grow rapidly, as did remittance inflows, and net inflows of foreign portfolio and direct investment were positive. As a result, foreign exchange reserves grew to around US\$600 billion, equivalent to about 8 months of imports.

9. **The fiscal deficit has narrowed and public debt has declined.** The general government deficit is estimated to have declined to 9.0 percent of GDP in FY22/23, from 12.6 percent in FY20/21, and public debt to have fallen to 83 percent from 89.3 percent. Fiscal consolidation was driven by strong revenue growth, particularly in goods and services tax revenues, and the gradual withdrawal of most pandemic-related stimulus measures. Higher than expected revenues and lower recurrent spending allowed capital spending to grow, including investments in infrastructure designed to crowd-in private investment.

2.2. MACROECONOMIC OUTLOOK AND DEBT SUSTAINABILITY

10. **Growth is expected to moderate further to 6.3 percent in FY23/24, returning gradually to its potential level.** Domestic demand growth is expected to decelerate, reflecting fading base effects and renewed price pressures, and external demand is expected to remain subdued due to the global growth slowdown. However, public and private investment growth is expected to remain robust. Public investments will be driven by the GoI's infrastructure push and private investments by improved balance sheets of banks and firms. On the supply side,

² The steep decline in NPLs is attributable to (i) the one-time restructuring measures by RBI for both corporate and MSME accounts to mitigate the impact of Covid-19, (ii) improvement in the performance of state owned banks and (iii) tighter credit appraisal standards and supervision after COVID.



contact-intensive services are expected to recover to pre-pandemic levels, and strong expansion in construction should boost industrial growth. Agricultural growth is expected to moderate due to deficient monsoon rainfall.

11. **Headline inflation is projected around 5.9 percent in FY23/24, reflecting the temporary spike in Q1.** Over the remainder of the year, inflationary pressure from food prices is expected to abate gradually as domestic supply normalizes. Core inflation should remain modest, around 4.5 percent. The RBI has signaled its intention to raise rates if inflationary pressures become more broad-based.

12. **The CAD is expected to narrow to 1.4 percent in FY23/24 and continue to be adequately financed by stable investment flows.** Weak global growth will weigh on merchandise exports, but softening commodity prices will help reduce imports. The goods trade deficit will be partially offset by growing services exports and stable remittances. Overall, the CAD is expected to be adequately financed by foreign capital flows and India's reserves position to remain robust at over eight months of import cover.

13. **In the FY23/24 Union Budget, the government reaffirmed its commitment to fiscal consolidation and reorientating public spending toward investment.** The FY23/24 budget projects further consolidation in the central government's fiscal deficit to 5.9 percent of GDP³, from 6.5 percent in FY22/23, with the share of capital spending projected to increase from 14 percent of total spending to about 19 percent. The budget deficit and debt repayments will largely be financed through domestic bond issuances⁴. In net terms, the Union budget envisages domestic borrowings of 5.8 percent of GDP (compared with 6.3 percent in FY22/23) and external borrowing of about 0.1 percent of GDP (unchanged from FY22/23).

14. **The public debt-to-GDP ratio is high but stable and sustainable.** Under the baseline scenario, the debt-to-GDP ratio stabilizes around 82-83 percent over the next three years. In a combined macro-fiscal shock scenario, the debt-to-GDP ratio would increase sharply to over 95 percent. However, debt sustainability risks are mitigated by the structure of India's public debt which is: (a) mostly domestically issued (only around 3 percent of public debt is external); (b) almost entirely held by residents; and (c) of medium- or long-term maturity⁵. The composition of debt is unlikely to change significantly over the projection period and the bulk of the financing needs will be met by domestic market issuances of medium- and long-term debt.

15. **Risks include slower-than-expected global growth, higher oil prices, and more persistent inflationary pressures.** Weaker global growth would weigh on external demand for India's exports. Higher oil prices would lead to an increase in the merchandise trade deficit and external financing needs. Higher oil prices and inflationary pressures from domestic food prices could lead to further hikes in the policy interest rate. However, the risks remain manageable given India's large and diversified economic base, and significant foreign reserve buffers.

16. **India's macroeconomic policy framework is adequate for development policy financing.** India's growth prospects are robust and the country has built and maintained a track record of sound fiscal and monetary policies. After relaxing their fiscal stance during COVID, the authorities have emphasized fiscal consolidation, with moderation in current spending growth to make space for higher public investment. In response to inflationary

³ The reduction in the fiscal deficit is expected to be driven by a 1.1 percentage point decline in recurrent spending with (i) the withdrawal of some pandemic-related measures (a 0.4 percentage point decline in food subsidies); (ii) a 0.2 percentage point decline in fertilizer subsidies (thanks to lower commodity prices on average); and, (iii) slower growth in recurrent spending relative to projected nominal GDP growth (10.5 percent), particularly on defense and central government pensions.

⁴ The government has projected gross market borrowing of over US\$200 billion and borrowing from other domestic sources such as the National Small Savings Fund and State Provident Funds, of about US\$66 billion. In addition, the government has budgeted for gross external borrowing of US\$ 10.4 billion, of which US\$6.6 billion would be from IFIs.

⁵ According to the IMF Fiscal Monitor, in 2022, average term to maturity for government securities was 9.8 years, compared with an average of 7.5 for emerging market countries.



pressures, the Reserve Bank of India (RBI) has frontloaded interest rate hikes and reaffirmed its commitment to price stability. India is not immune to adverse external financial developments or terms-of-trade shocks, but high reserves levels, a floating exchange rate, and limited external financing needs provide adequate buffers. Public debt is high but sustainable and relatively resilient to different shocks.

2.3. IMF RELATIONS

17. **The International Monetary Fund (IMF) does not have an active program in India but it carries out supervision and Article IV consultations annually.** The most recent Article IV Consultation Report was completed on December 23, 2022⁶. It commended the Indian authorities for responding to commodity price increases and elevated inflation by front-loading monetary policy tightening and introducing fiscal policy measures to support vulnerable groups. It highlighted opportunities for further structural reforms, including to address the adverse impact of climate change to secure strong and sustainable growth.

3. GOVERNMENT PROGRAM

See Annex 6

4. PROPOSED OPERATION

4.1. LINK TO GOVERNMENT PROGRAM AND OPERATION DESCRIPTION

4.1. PROGRAM DEVELOPMENT OBJECTIVES

18. **The Program Development Objective (PDO) of the proposed DPO is to accelerate the development of low-carbon energy in India.** To achieve the PDO, the operation supports the government's efforts to (i) develop a GH industry and ecosystem; (ii) scale-up RE; and (iii) enhance climate finance for low-carbon energy investments. The three pillars are interlinked since additional RE is critical to making GH viable, and significant additional financing will be needed to meet the large investment needs of GH and RE.

19. **The operation is aligned with the government's own program that reflects a strategic bet on GH and a major push on boosting RE penetration.** Through the NGHM, the government is pursuing three avenues to develop GH, by: (i) creating demand and demonstrating viability through pilot projects; (ii) incentivizing domestic production of electrolyzers and GH; and (iii) building an enabling ecosystem for GH markets. By 2030, the NGHM is targeting: (a) 5 million tons of GH produced annually; (b) 125 GW of additional RE; and (c) leveraging US\$100 billion of private investments. The GoI has already adopted policies to boost investment in RE⁷ and improve RE grid integration⁸. In FY23, the budget provided viability gap funding for 4 Gigawatt-hour (GWh) Battery Energy Storage Systems (BESS)

⁶ <https://www.imf.org/en/News/Articles/2022/12/19/pr22444-india-imf-executive-board-concludes-2022-article-iv-consultation>

⁷ (a) fiscal incentives that have brought down solar PV costs; (b) a waiver of inter-state transmission system charges for RE power generators; (c) support for large-scale grid-connected solar parks; (d) RPOs for distribution utilities; (e) must-run status for RE power plants to ensure grid integration; (f) production-linked incentives for solar cells and modules; (g) the KUSUM scheme to promote solar pumps in rural areas; and (h) incentives to adopt electric and hybrid vehicles through the FAME Scheme

⁸ including via the National Battery Storage and Transformative e-Mobility Mission, expanding transmission networks, and increasing power generation flexibility



and the Central Electricity Authority (CEA) issued a comprehensive Transmission Plan for integrating 500 GW of RE into India's power grid by 2030.⁹ Finally, the government has sought to stimulate private sector investments in RE and GH through direct financial incentives¹⁰ and policy reforms¹¹. India has also launched a Carbon Credit Trading Scheme (CCTS) to accelerate and enhance the energy transition¹². The proposed operation seeks to support and leverage these complementary government initiatives.

20. **Climate Co-benefits:** The program supported by this DPO will help scale up investments in greening the energy mix and accelerate India's low-carbon transition. Each pillar contributes to India's climate change mitigation agenda.

4.2. PRIOR ACTIONS (PA), RESULTS AND ANALYTICAL UNDERPINNINGS

21. **All but two prior actions -PAs #2 and #3 have been met.** With respect to PA#2 the government has taken steps to encourage "demand sectors" to use GH but these are non-binding and insufficient to guarantee offtake. Prior Action #3 is in progress to the extent that the relevant guidelines have been prepared, but the consultation and approval process may take more time than expected.

Pillar 1: Promoting green hydrogen¹³.

22. **GH remains a frontier technology, the viability of which will hinge on developing the market to a critical scale; in turn, scaling up requires coordinated policy and regulatory reforms to promote production and demand.** Under DPO1, the Ministry of New and Renewable Energy (MNRE) approved the NGHM, that provides an overarching framework for mobilizing private sector investments and creating market demand for GH (PA 1 DPF1). PAs 1-4 under the proposed operation seek to operationalize the NGHM through measures to bridge the viability gap for GH (by incentivizing production and market demand), putting in place a conducive regulatory environment, and mobilizing private investment.

23. **PA #1: Investments in GH are currently too costly and risky for the private sector to take the first steps alone; to address this constraint, the MNRE has issued two incentive schemes.** The first scheme targets GH production with an incentive of INR50 (\$0.625)/kg of GH (about 15 percent of GH production costs), which would phase out over the next three years. The second scheme targets electrolyser manufacturing, with an incentive of INR4,440 (\$55)/kW (about 8 percent of electrolyser production costs), also to phase out over time. These incentive schemes aim to improve cost competitiveness of GH and its derivatives and encourage their large-scale utilization and hence attract private investments. They will be implemented by the Solar Energy Corporation of India Limited (SECI) and allocated according to a competitive process for selecting beneficiaries and specific eligibility criteria.

⁹ CEA's Transmission System for Integration of over 500 GW RE Capacity by 2030 (https://cea.nic.in/wp-content/uploads/notification/2022/12/CEA_Tx_Plan_for_500GW_Non_fossil_capacity_by_2030.pdf)

¹⁰ Production-Linked Incentives for solar PV and BESS, and viability gap funding for 4 GWh of RE, as well as US\$2.4 billion including incentives for GH

¹¹ the GoI has adopted a framework for the issuance of sovereign green bonds and created a separate reserve/corpus fund titled Sovereign Green Fund in the public accounts

¹² The SEBI has issued circulars for Green Debt Securities to strengthen the disclosure requirements for issuance and listing of green debt securities

¹³ Analytical underpinnings for Pillar 1:

WB: *Green Hydrogen Opportunities and Roadmap for India* (2023)

WB: *Geographic Information System mapping of demand and supply centers for green hydrogen hubs in India* (2023)

WB: *Developing financing mechanisms to bridge the cost gap between grey and green hydrogen in India* (2023)

MoEFCC: *India's Long-Term Low-Carbon Development Strategy* (2022)

WB: *Support to MNRE in developing GH Hub Policy* (ongoing)

WB: *Study on Desalination of Water and its use in GH* (ongoing)



Further, SECI has launched the two tenders under the incentive schemes. *[Status: met – the incentive schemes for electrolyzer manufacturing and GH production were issued on June 28, 2023 by MNRE]*

24. **PA #2: Additional investments in GH supply will not materialize in the absence of guaranteed demand (offtake); the government intends to promote GH offtake in select industries to initiate this virtuous cycle of demand and supply and eventually reach commercially viable scale.** However, this is an area where further progress remains to be achieved. The MNRE has issued inter-governmental letters to the Ministries of Petroleum and Natural Gas (for refineries), Fertilizers¹⁴, Road, Transport and Highways, Steel, and, Ports, Shipping and Waterways to request them to share their respective trajectories for the demand of GH and derivatives in their respective sectors, but these are non-binding. The team is exploring alternative options through direct engagements with the demand-side ministries.¹⁵ *[Status: not met. The prior action needs to be revised, as it currently lacks specificity and the inter-ministrial letters issued by MNRE have no binding value].*

25. **PA #3: This is an important measure to increase domestic uptake of GH since bulk procurement of GH and derivatives would contribute to mitigate off-taker and demand risks; demand aggregation will provide more certainty to producers, generate economies of scale, and help optimize production and transportation.** Guidelines for the bidding process that ensures transparency and maximizes value-for-money will be critical. MNRE already has a draft available and is in consultations with the various relevant ministries to aggregate the demand for GH and derivatives.¹⁶ *[Status: not met. The process is ongoing but it is unclear whether it will conclude by October given the multiple layers of consultations and approvals that are required before the NGHM Empowered Group can examine and notify the new guidelines].*

26. **PA #4: The the MNRE has introduced the Green Hydrogen Standard to clearly define what constitutes “green hydrogen”.** The proposed incentives scheme and bulk procurement will require a clear definition of what constitutes “green hydrogen” under the NGHM. To provide such clarity, the MNRE has introduced the Green Hydrogen Standard that requires less than 2 kg CO₂/kg generated during the production of hydrogen to be considered green. A detailed methodology for measurement, reporting, monitoring, onsite verification and certification of GH and its derivatives will be issued by MNRE. The Bureau of Energy Efficiency (BEE) has been designated as nodal authority for accrediting agencies for monitoring, verifying and certifying GH production projects. This is another key policy to ensure the successful implementation of the NGHM, as GH projects must meet the GH standards regarding RE use to be eligible for government incentives. *[Status: met. Green Hydrogen Standards for India have been issued on August 18, 2023 by MNRE]*

27. **Expected results:** Pillar one is expected to kickstart domestic GH and electrolyzer production, targeted to incentivize reach 450,000 metric tons per annum and 1.5 GW, respectively, by 2026. These revised targets are based on the recently issued tender documents by SECI under the Incentive Schemes. Further bids will be issued based on the outcome of the these tenders to facilitate targets for 2030 as set forth under NGHM.

¹⁴ In 2021, India was importing urea, di-ammonium phosphate, and ammonia, for a total amount of US\$ 6 billion.

¹⁵ The World Bank is supporting MNRE in developing its GH Hub Policy where the early investments are likely to come. This shall also build on the geospatial mapping of GH hubs which the World Bank is undertaking. The World Bank has also initiated the work on studying desalination of water and its requirements in the context of GH production.

¹⁶ The World Bank is supporting the MNRE in developing its GH Hub Policy that would allow for demand aggregation as well.



Pillar 2: Scaling up renewable energy¹⁷.

28. **Pillar two seeks to catalyze increases in the share of RE in the energy mix.** Under DPO1, the government took measures to mandate increased use of RE by distribution utilities (PA 4 DPO1), remunerate the provision of storage services (PA 5 DPFO1), incentivize a scale up in RE generation capacity (PAs 6 and 7 DPO1), and boost the production of solar PV (PA 8 DPO1). DPO2 seeks to extend these measures by ensuring RE sources are better integrated into the grid. Because of RE's inherent variability, RE grid integration needs to be supported, including through energy storage.

29. **PA #5: The Electricity Grid Code has been amended to improve RE grid integration.** Due to the variable nature of RE, grid integration is a significant issue. Technical remedies are typically defined in the electricity grid code of a country. India amended its Electricity Grid Code and updated the technical and commercial rules for power market participants that are connected to the inter-state transmission system (ISTS). The following features have been added: (a) a resource planning code that supports demand forecasting, generation resource planning, and energy storage systems; (b) a newly-added protection code, which ensures stability of the power system; (c) an operating code that promotes availability of system reserves and frequency control mechanisms such as ancillary services (to ensure reliability), thereby providing for inertia (to provide system support in the event of power loss due to a failed generator) and grid stability; and (d) a schedule and dispatch code introducing a framework for nationwide optimization of security-constrained economic dispatch¹⁸ by generator in the real-time market. *[Status – met. The IEGC was published on July 11, 2023 and will be effective from October 1, 2023].*

30. **PA #6: A new prior action on policy support for Battery energy storage systems (BESS) is proposed.** BESS are critical for RE grid integration. To reduce BESS costs, Cabinet has approved a policy providing a financial incentive of up to 40 percent of the capital costs of BESS as viability gap fund for 4000 MWh of BESS by 2030-31. This is a watershed policy to help reduce the costs of BESS and attract private sector investments. Once the costs of RE and BESS reach cost parity with coal power plants, there will be no need to build new coal power plants to meet the growing power demand. *[Status – met. The Scheme was approved by India's Cabinet on September 6, 2023]*

31. **The actions under the second pillar of the series are expected to result in:** a significant increase in the share of RE in power consumption (from 25 to 33 percent between 2022 and 2026) and the share of solar and wind power consumed through energy storage in total power consumption (from 0 to 2 percent over the same period).

Pillar 3: Enhancing climate finance for low-carbon energy investments¹⁹.

32. **Pillar three aims to stimulate climate financing for private investments in GH and low-carbon energy.** Under the first DPO the government had put in place the legal framework for carbon trading (PA 9 DPF1) and upgraded the regulatory framework for green debt issuances (PAs 10-11 DPF1). DPO 2 proposes further reforms in

¹⁷ Analytical underpinnings for Pillar 2:

WB: *Battery Storage Assessment at Intra-State Transmission and Distribution Network* (2021)

WB: *A Roadmap for Developing Wholesale Power Market in India* (2021)

MoP: *Report of the Group on Development of Electricity Market in India* (2023)

WB: *Report on India Energy Transition to meet Net Zero 2070* (2023)

MoP: *National Framework for Promoting Energy Storage Systems* (2023)

¹⁸ Security-constrained economic dispatch is a mathematical model to generate the most economic generation dispatch while considering key system operation constraints, such as power balance constraint, reserve requirement constraints, transmission security constraints, as well as generation limitations, such as ramp rates, minimum and maximum output.

¹⁹ Analytical underpinnings for Pillar 3:

WB: Technical assistance under *Private and Financial Sector Equitable Growth*

WB: Designing carbon market in India through technical assistance on partnership for market readiness



both areas.

33. **PA #7: The CCTS²⁰ was issued to enable a more efficient and market-based low-carbon energy transition.** The existing Perform-Achieve-Trade (PAT) scheme, in place since 2012, allows institutions to trade to achieve energy performance benchmarks. The amended legislation provides fungibility of the PAT certificates to carbon certificates and creates a domestic carbon market that goes beyond the entities covered by PAT scheme by allowing the certificates to be used by companies not covered under the PAT scheme for their voluntary commitments in a phased approach. *[Status – met. The CCTS was issued by Ministry of Power on June 28, 2023]*

34. **PA #8: Enhancing the flow of green finance while preventing greenwashing requires an enabling regulatory framework on Environmental, Social and Governance (ESG) ratings, disclosure and labelled products.** In July 2023, the Securities and Exchange Board of India (SEBI) introduced several complementary regulatory guidelines on ESG ratings, disclosures and funds, which are expected to significantly improve the allocation and monitoring of green finance instruments and funds in India. First, it introduced a comprehensive regulatory framework for ESG rating providers to address the concern that previously unregulated ESG ratings had limited comparability and could lead to greenwashing. ESG rating providers are now required to register with SEBI and disclose their rating methodologies. SEBI also introduced more stringent ESG disclosure requirements for the largest listed firms as well as external validation requirements. Second, SEBI introduced the Business Responsibility and Sustainability Reporting (BRSR) Core reporting format. The BRSR Core format requires the largest listed entities to ensure reasonable assurance through external verification of data reported on nine key areas, including GHG emissions and energy use. These large firms will also be required to make similar disclosures for their value chain, covering at a minimum 75 percent of their total purchases or sales. Third, to enable differentiated strategies in ESG investments, SEBI allowed mutual funds to introduce multiple ESG themed funds, compared to a single ESG fund previously. Also, the investment guidelines for ESG funds have been aligned with BRSR Core and other guidelines. *[Status – met: SEBI has issued the regulatory framework for ESG rating providers in July 2023]*

35. **Pillar three is expected to result in** the launch and operationalization of a carbon trading market with specified GHG emissions targets. The impact will be assessed by looking at additional cumulative issuances of onshore green debt securities that are expected to increase by at least 30 percent between 2022 and 2026 from a 45.4 billion baseline, and an increase in the volume of sovereign green bonds issued from 2.3 billion in 2022 to 6 billion in 2026.

4.3. LINK TO CPF, OTHER BANK OPERATIONS AND THE WBG STRATEGY

See Annex 6

4.4. CONSULTATIONS AND COLLABORATION WITH DEVELOPMENT PARTNERS

See below

5. OTHER DESIGN AND APPRAISAL ISSUES

²⁰ Formally known as the Indian Carbon Market.



36. **The DPO is not expected to have direct negative impacts on poverty.** No prior action is likely to have a direct negative impact on poverty. However, the transition from brown to renewable energy and from grey to green hydrogen will create winners and losers. New employment opportunities will be created in the RE and GH space and secondary labor markets. At the same time, localized employment losses may occur in the transition away from grey hydrogen and fossil-based industries (although not as a direct result of any PA). In the longer-term, the transition to cleaner sources of energy can support welfare and equity gains, via health co-benefits, increased worker productivity, and reduced exposure and vulnerability to climate and environmental hazards.

37. **GH production also entails substantial risks related to land acquisition, involuntary resettlement, occupational and community health and safety.** This is particularly true since it makes economic sense to locate GH production facilities next to demand centers. While no priori action directly targets the building of or choice of location for GH plants, the supported incentives programs (PAs #1-4) will result in expanded GH production. To help mitigate these risks, the World Bank is in an active dialogue with the relevant institutions to advance a TA program²¹ for the proposed DPO to develop relevant skills to facilitate employment transition to green jobs and to support desalination of water that can be used for GH production.

38. **The reforms supported through this DPO -all prior actions taken together- are likely, overall, to have a positive impact on India's environment, forests and other natural resources,** due to lower GHG emissions and co-benefits for public health. However, GH production facilities may have a negative localized impact on freshwater availability.

39. **This operation aligns with the mitigation and adaptation and resilience goals of the Paris Agreement.** The DPO reform program is consistent with the country's climate commitments and the Bank's climate analysis, including findings from the CCDD under preparation. Regarding mitigation goals, all PAs seek to remove significant and persistent barriers to the country's transition to a low-GHG emissions development pathway. Regarding adaptation and resilience goals, the contributions of the PAs to the Project Development Objective (PDO) are unlikely to be at risk from climate hazards.

40. **Consultations.** The Bank team has been consulting closely with MNRE, the nodal agency for NGHM as well as with the various stakeholders in the context of GH and the broader context of DPO1 and DPO2. Further, MNRE regularly carries out consultations with private sector stakeholders (RE developers and GH technology providers) and thinktanks to advance the interventions proposed under NGHM. The key insight from stakeholder engagements is that the current high costs of GH is a major constraint. The MNRE and the Bank team have also coordinated closely with the United Kingdom's Foreign, Commonwealth, and Development Office that has extended a guarantee for a principal amount of US\$1 billion under the first operation as well as with other development partners active in the GH space in India. Finally, the Bank and the Asian Infrastructure Investment Bank are exploring co-financing of DPO2. International Finance Corporation is actively consulting with potential investors in the GH space.

41. **Fiduciary arrangements for the DPO are satisfactory, including public financial management (PFM) systems, procurement systems and the foreign exchange control environment.** Assessments conducted by independent thinktanks and the Bank conclude that country fiduciary risk is moderate at the country level, including in planning, budgeting, financial reporting, and public investment management. Public procurement assessed to be functioning moderately well, with islands of excellence. Over 80 percent of procurement is carried out through open

²¹ The Ministry of Skill Development and Entrepreneurship has developed initiatives targeting engineers and researchers to better understand the basics of GH production. The Bank will help design the detailed programs educating engineers and technicians on design, engineering, operations and maintenance, and storage and transportation, and on-the-ground training on safe handling of hydrogen supply and end-use systems. The World Bank is either providing or is in advance dialogue to provide TA to bring in global best practices on GH production and use and mitigating the environmental and social risks to build institutional capacity both at the federal and state level.



competition procedures and e-procurement is mandatory for all procurement valued above INR 5 lakhs. The Bank has reasonable assurance that the control environment for foreign exchange in the RBI is satisfactory for the purposes of this DPO.

5.1. POVERTY AND SOCIAL IMPACT

See above

5.2. ENVIRONMENTAL, FORESTS, AND OTHER NATURAL RESOURCE ASPECTS

See above

5.3. PFM, DISBURSEMENT AND AUDITING ASPECTS

See above

5.4. MONITORING, EVALUATION AND ACCOUNTABILITY

Not applicable at PCN stage

6. SUMMARY OF RISKS AND MITIGATION

42. **The overall residual risk rating for the operation is substantial.** Table 1 presents the residual risk ratings by category. The categories of "Sector Strategies and Policies", "Technical Design", "Environmental and social" and "Stakeholders" are rated substantial risks to the achievement of the PDO. A description of these risks and their respective mitigation measures is provided below.

43. **Sector strategies and policies risk is substantial.** GH is still at a nascent stage and its development involves multiple ministries and stakeholders. The proposed DPO will serve as a platform to coordinate across these multiples stakeholders but residual risks remain. NGHM provides a solid foundation to address some of the sectoral risks.

44. **Technical design risk is substantial.** There are still significant uncertainties around GH technologies and costs, electrolyzer manufacturing capacities, and storage and transportation of hydrogen. Given India's clear ambition to become a pioneer in GH and the potential sizeable payoffs for India and globally, it makes sense for for the GOI to absorb the front-end risks associated with a nascent yet promising technology of GH and for the World Bank to accept significant residual risks to the achievement of the development objectives. The World Bank is working with key public sector players such as Steel Authority of India Limited (SAIL) to explore (Green) Hydrogen Direct Reduction (H2DRI) technology. This will help ensure that the risks are assessed upfront and appropriate mitigation measures are factored in investment decisions.

45. **Environmental and social risks are substantial.**²² Some of the PAs may have significant social implications linked to RE and GH development, such as land acquisition resulting in loss of livelihood, involuntary resettlement,

²² There are allegations of forced labor in the production of solar panels and components. This DPO focuses on policies and institutional reforms in India. DPO proceeds are not earmarked to any specific purpose such as the manufacturing or procurement of solar panels or components.



labor issues, occupational and community health and safety issues, and reduced freshwater availability. Large manufacturing activities are likely to have environmental and ecological impacts depending on the location of such activities, and GH and other RE sources (solar, wind, offshore wind) are only partly covered by the existing legal and regulatory frameworks on minimizing or avoiding impacts on natural resources, environment, and biodiversity. The World Bank has done a geospatial mapping of GH hubs that avoids the sensitive areas such as protected forests, critical zones for underground water, land use land cover layers amongst others. The World Bank is also working on an assessment of desalination of water and its possible applications in the context of GH and derivatives, including disposal of brine. Further, the World Bank is in advanced discussions with multiple institutions to enhance the skills in this sector.

46. **Stakeholders risk rating is substantial.** During consultations, stakeholders such as policy makers, RE developers, electrolyzer manufacturers, GH producers, GH users, industries, and gas pipeline companies, at both central and state levels, highlighted the high costs of GH as a major constraint. The proposed operation supports government programs which will provide incentives to bring down the costs of GH to mitigate this risk. In the longer term, as the market expands and the production ramps-up, thanks to the reforms program, costs will decline even further. Also, the respective ministries and states are formulating their strategies to promote green technologies in their respective area of jurisdictions along with introducing fiscal incentives that shall contribute to bring down the gap between grey hydrogen and GH.

Risk Categories	Rating
1. Political and Governance	● Moderate
2. Macroeconomic	● Moderate
3. Sector Strategies and Policies	● Substantial
4. Technical Design of Project or Program	● Substantial
5. Institutional Capacity for Implementation and Sustainability	● Moderate
6. Fiduciary	● Moderate
7. Environment and Social	● Substantial
8. Stakeholders	● Substantial
9. Other	
Overall	● Substantial





ANNEX 1: POLICY AND RESULTS MATRIX

Prior Actions		Results			
Prior Actions under DPF 1	Prior Actions under DPF 2	Indicator Name	Baseline (June 30, 2023)	Current	Target (June 30, 2026)
Pillar 1: Promoting GH					
PA #1: The Borrower, through its Union Cabinet, has approved the NGHMs, which will facilitate demand and includes incentives for electrolyzer manufacturing, and GH production.	PA #1: The Borrower shall: (i) notify the incentive schemes for GH production and electrolyzer manufacturing, which define the detailed incentives, specify the eligibility criteria for recipients, and outline the framework for selection through a transparent and competitive process, and (ii) appoint the implementing agency(ies) which will manage the incentive schemes.	Results Indicator #1: GH production capacity incentivized (tonnes)	0	0	450,000
	PA #2: The Borrower shall select certain sectors and promote the use of GH/its derivatives in these sectors. PA #3: The Borrower, through the MNRE, shall issue a guideline for transparent and competitive bidding for bulk procurement of GH and its derivatives. PA #4: The Borrower shall notify a GH standard that defines the eligibility criteria for consideration as “green” hydrogen.	Results Indicator #2: Domestic electrolyzer manufacturing capacity incentivized (MW)	0	0	500
PA #2: The Borrower, through the MNRE, has issued recommendations to the		Results Indicator #3: Number of additional GH safety	0	0	10



Prior Actions		Results			
relevant ministries on the adoption of relevant regulations, standards, codes, best practices, and procedures related to safety aspects of: (a) GH production; (b) GH safe storage and handling; and (c) GH utilization in the mobility sector.		standards notified			
PA #3: The Borrower, through the Ministry of Power (MoP), has notified extension of the waiver of the ISTS charges towards consumption of RE for GH and its derivatives production for projects to be commissioned until December 31, 2030.					
Pillar 2: Scaling up RE					
PA #4: The Borrower, through the MoP, has issued a government order on: (a) renewable purchase obligations (RPO); and (b) energy storage obligations that mandate the purchase of a minimum specified share of power consumption from RE and energy storage, respectively, for all electricity distribution utilities.	PA #5: The Borrower, through the Central Electricity Regulatory Commission (CERC), shall issue the amended Indian Electricity Grid Code regulation to improve RE grid integration	Results Indicator #4: Share of RE in power consumption (percentage)	25	25	33
		Results Indicator #5: Share of solar/wind with/through energy storage in total power consumption (percentage)	0	0	2
PA #5: The Borrower: (a) through the CERC, has issued and notified the CERC (Ancillary Services) Regulations 2022, that provide a market-based mechanism to remunerate provision of energy storage services, and (b) through the MoP, has issued the guidelines on pumped storage hydropower.	PA #6: To reduce BESS costs, Cabinet has approved a policy providing a financial incentive of up to 40 percent of the capital costs of BESS as viability gap fund for 4000 MWh of BESS by 2030-31.	Results Indicator #6: BESS incentivized (MWh)	0	0	TBA
PA #6: The Borrower, through the MNRE, has issued an implementation regulation		Result Indicator #7: Additional bids for RE capacity	0	0	75



Prior Actions		Results			
to guide the bidding of 50 GW of RE capacity, including solar photovoltaic, wind, and firm power from RE and energy storage, each year from FY23-24 to FY27-28.		issued (GW)			
		Result Indicator #8: GHG emissions avoided (million tonnes per annum)	0	0	40
PA #7: The Borrower: (a) through the MNRE, has adopted a strategy for competitive bidding for allocation of offshore wind sites; and (b) through the MoP, has extended the waiver of the ISTS charges toward offshore wind until December 31, 2032.		Result indicator #9: Offshore wind sites awarded (equivalent to power capacity GW)	0	0	4
PA #8: The Borrower, through the MNRE, has issued an updated policy to provide PLIs to high-efficiency solar PV to remove the supply chain bottlenecks and reduce the costs of solar PV.		Results Indicator #10: Capacity for domestic manufacturing of high-efficiency solar PV cells and modules added (GW)	0	0	48
Pillar 3: Enhancing climate finance for low-carbon energy investments					
PA #9: The Borrower: (a) through the Parliament, has approved amendments to the Energy Conservation Act, 2001 that provide the legal framework for the launch of a national carbon market; and (b) through the Ministry of Environment, Forest and Climate Change, has approved GHG mitigation activities, including GH, green ammonia, RE with storage, and offshore wind to be eligible for international carbon market to mobilize international financing under the Paris Agreement.	PA #7: The Borrower, through the MoP, shall issue a national CCTS notification and regulation that specifies the governance structure, trading rules, and regulations.	Results Indicator #11: Launch of a national carbon market	No carbon market exists	CCTS issued	GHG emission intensity targets allocated and carbon trading operational



Prior Actions		Results			
PA #10: The Borrower, through the SEBI, has issued amendments to the existing regulatory framework for green debt security issuance, to specify the taxonomy and definition of green debt securities to include the emerging areas of sustainable financing and improve the information disclosure standards.	PA #8: The Borrower, through the SEBI, has issued regulatory guidelines on ESG rating, disclosures, and ESG themed funds, to support higher levels on transparency and comparability to enable flow of green finance and avoid greenwashing.	Result Indicator #12: Cumulative additional issuance of onshore green debt securities (corresponding to 30% increase from 45.4 bn baseline)	0	0	13.6 bn
PA #11: The Borrower, through the Ministry of Finance, has issued a transparent Sovereign Green Bond Framework for evaluating, selecting, and financing green public spending, including on GH, solar and wind energy, pursuant to which a separate reserve/corpus fund titled 'Sovereign Green Fund' has been created.		Results Indicator #13: Sovereign green bonds issued (cumulative) (US\$ billion)	2.3	2.3	6.0



ANNEX 2: FUND RELATIONS ANNEX

TBA



ANNEX 3: LETTER OF DEVELOPMENT POLICY

TBA



ANNEX 4: ENVIRONMENT AND POVERTY/SOCIAL ANALYSIS TABLE

TBA



ANNEX 5: MACROECONOMIC TABLES

Table A2.1. Selected Economic and Financial Indicators

	FY19/20	FY20/21	FY21/22	FY22/23	FY23/24	FY24/25	FY25/26
<i>Annual percentage change, unless otherwise indicated</i>							
National Accounts							
GDP at constant prices (%)	3.9	-5.8	9.1	7.2	6.3	6.4	6.5
Consumption	5.0	-4.6	10.5	6.4	5.6	5.9	6.3
Investment	1.1	-7.3	14.6	11.4	8.9	7.8	7.3
Exports of goods and services	-3.4	-9.1	29.3	13.6	0.9	6.7	8.2
Imports of goods and services	-0.8	-13.7	21.8	17.1	3.0	7.2	8.7
<i>Sectoral contribution to growth</i>							
Agriculture (ppts)	0.8	0.6	0.5	0.6	0.5	0.5	0.5
Industry (ppts)	-0.4	-0.2	3.3	1.3	1.6	1.8	1.8
Services (ppts)	3.2	-4.1	4.3	4.6	3.7	3.7	3.7
Inflation							
GDP deflator	2.4	4.7	8.5	8.2	5.0	4.6	4.1
Consumer prices (average)	4.8	6.2	5.5	6.7	5.9	4.7	4.1
Selected Monetary Accounts							
Banks' credit	6.1	5.6	8.6	15.0	-	-	-
Broad money (M2)	11.1	16.1	10.7	6.9			
External sector							
Exports fob	-2.1	-5.9	36.1	14.3	2.8	9.7	10.5
Imports fob	-5.8	-14.9	48.4	18.0	1.6	8.9	11.8
Terms of trade	7.2	5.9	-14.3	-0.2	3.3	1.2	-0.7
<i>Percent of GDP, unless otherwise indicated</i>							
Current account balance (incl. grants)	-0.9	0.9	-1.2	-2.0	-1.4	-1.2	-1.6
Foreign Direct Investment	-1.5	-1.6	-1.2	-0.8	-1.1	-1.4	-1.5
Debt							
Public debt (external and domestic)	73.6	89.3	84.8	82.9	82.9	82.5	82.4
External government debt	2.9	3.3	2.8	3.1	2.9	2.7	2.5
Debt service	4.8	5.3	5.2	5.1	5.4	5.4	5.4
Fiscal Accounts							
Total revenue and grants	19.7	18.5	19.5	18.9	19.5	19.9	20.0
Total expenditure and net lending	26.9	31.1	29.1	27.9	28.1	27.9	27.9
Overall fiscal balance (with grants)	-7.2	-12.6	-9.6	-9.0	-8.7	-8.1	-7.9
Memorandum items							
GDP per capita (%)	2.8	-6.7	8.2	6.5	5.4	5.5	5.5
Gross reserves (US\$ millions, EOP)	477806.8	576983.9	607309.1	578449.2	609066.5	651047.0	711093.0
In months of next year's imports)	11.1	9.0	8.1	7.6	7.3	7.0	-
Nominal GDP (INR, trillions)	201.03	198.30	234.71	272.40	304.05	338.59	375.17
Nominal GDP (US\$, billions)	2836.3	2672.5	3149.9	3388.5	3730.7	4085.7	4459.9

Source: MoSPI, RBI, Ministry of Finance (MoF), CEIC and WB staff estimates

Notes: FY refers to fiscal year (April-March)



Table A2.2. Key Fiscal Indicators (Percent of GDP)

	FY19/20	FY20/21	FY21/22	FY22/23	FY23/24	FY24/25	FY25/26
Total revenue and grants	19.7	18.5	19.5	18.9	19.5	19.9	20.0
Tax revenue	16.1	15.8	16.6	16.5	16.9	17.2	17.2
Direct taxes	6.0	5.6	6.0	6.1	6.4	6.5	6.4
Indirect taxes	10.0	10.1	10.6	10.3	10.5	10.6	10.7
Taxes on goods and services	9.5	9.4	9.7	9.5	9.7	9.8	9.9
Taxes on international trade	0.5	0.7	0.9	0.8	0.8	0.8	0.8
Non-tax revenue	3.1	2.5	2.7	2.3	2.3	2.4	2.4
Other Revenues (disinvestment)	0.5	0.3	0.2	0.2	0.3	0.3	0.4
Total expenditures	26.9	31.1	29.1	27.9	28.1	27.9	27.9
<i>Current expenditures</i>	22.7	26.2	23.7	22.4	22.5	22.3	22.2
Wages and salaries	1.1	1.3	1.1	0.9	1.0	1.1	1.2
Goods and services	16.8	19.6	17.4	16.3	16.1	15.8	15.6
Interest on debt	4.8	5.3	5.2	5.1	5.4	5.4	5.4
<i>Capital expenditure</i>	3.8	4.5	5.0	5.1	5.3	5.3	5.3
Overall fiscal balance (incl. grants)	-7.2	-12.6	-9.6	-9.0	-8.7	-8.1	-7.9
Primary balance	-2.4	-7.2	-4.4	-3.9	-3.3	-2.7	-2.5
Financing	7.2	12.6	9.6	9.0	8.7	8.1	7.9

Source: MoSPI, RBI, MoF, CEIC and World Bank staff estimates



Table A2.3. Central Government and Government Guaranteed Debt Stock and Debt Service

	Debt Stock (end of period)			Debt Service			
	FY22/23			FY23/24	FY24/25	FY23/24	FY24/25
	in million US\$	Percent of Total	Percent of GDP	in million US\$		4	5
						(Percent of GDP)	
Total Debt	1892892	100.0	57.1				
Domestic Debt	1801782	95.2	54.4				
Treasury bills	100160	5.3	3.0	100160		3.0	
Treasury Bonds	1115076	58.9	33.6	53561	49822	1.6	1.3
Others	586547	31.0	17.7				
External Debt	91110	4.8	2.7	8700	9600	0.3	0.3
Multilateral creditors	63500	3.4	1.9	6400	7500	0.2	0.2
Bilateral Creditors	27600	1.5	0.8	2300	2100	0.1	0.1
Paris Club	27600	1.5	0.8	2300	2100	0.1	0.1
Non-Paris Club	0	0	0.0	0	0	0	0
Commercial Creditors	0	0	0.0	0	0	0	0
Memo items							
SOE guaranteed external debt	11000	0.6	0.0				
External Arrears	-	-	-				
INR per USD: Official (EOP)	82.2						

Source: MOF, World Bank staff calculations

Note: Based on available data for the central government; consolidated general government data has not been published

Table A2.4. Financing Needs (percent of GDP)

	FY19/20	FY20/21	FY21/22	FY22/23	FY23/24	FY24/25	FY25/26
1. Financing Requirements (i-ii)	0.9	-0.9	1.2	2.0	1.4	1.2	1.6
i. Current Account Deficit*	0.9	-0.9	1.2	2.0	1.4	1.2	1.6
ii. Net Errors and Omissions	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2. Financing Sources (i+ii)	0.9	-0.9	1.2	2.0	1.4	1.2	1.6
i. Capital Account Balance (a+b+c)	2.9	2.4	2.7	1.7	2.8	3.5	3.6
a. Net Foreign Direct Investment	1.5	1.6	1.2	0.8	1.1	1.4	1.5
b. Net Portfolio Investment	0.0	1.4	-0.5	-0.2	0.5	0.9	0.9
c. Net All Other Flows**	1.4	-0.6	2.0	1.1	1.2	1.2	1.2
ii. Change in reserve assets (1-2.i)	-2.1	-3.3	-1.5	0.3	-1.4	-2.3	-2.0
3. External Financing Gap (1-2)	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Source: RBI and staff calculations.

Note: *Includes merchandise and invisibles. **All other flows include short- and long-term debt flows (external assistance, commercial borrowings, and trade credits) and banking capital. Positive change in reserves indicates a decline.



ANNEX 6: LINK TO GOVERNMENT PROGRAM, COUNTRY PARTNERSHIP FRAMEWORK, OTHER BANK OPERATIONS, AND CORPORATE PRIORITIES

In addition to directly supporting the NGHM's implementation, this DPO is aligned with the government's RE missions and power market reforms, playing a key role in supporting the NGHM's implementation. The government has adopted a series of RE policies to mobilize significant private sector investments in RE and BESS—including fiscal incentives, RPOs, PLIs, waiver of ISTS charges, budget for viability gap funding—which have led to exponential growth of the RE market over the past decade. In the wake of this increasing share of RE in the power system, the government has also adopted a series of policies to improve RE grid integration—including the National Battery Storage and Transformative e-Mobility Mission, the expansion of transmission networks, and the introduction of more flexibility in power generation. Power market developments have also encouraged RE participation in the power markets, improved RE grid integration, and provided transparency on prices of RE. Finally, the government adopted a framework for the issuance of sovereign green bonds and created a separate reserve/corpus fund titled Sovereign Green Fund in the public account of India to provide complete transparency in terms of use of sovereign green bonds proceeds for financing green projects

The World Bank has been deeply engaged in GH in India over the past two years, through a series of analytical studies and technical assistance that have provided inputs to the National Green Hydrogen Roadmap, the NGHM, and the analytical underpinning for this proposed operation. The proposed operation is also aligned with the World Bank's current global engagement on GH. The World Bank launched a global Hydrogen-for-Development initiative (H4D) at the 27th Conference of the Parties to the United Nations Framework Conventions on Climate Change, which aims to provide global knowledge, technical assistance and financing for GH in the developing world. H4D aims to bring together the partner countries to enable capacity building, exchange knowledge on regulatory solutions, business models and technologies, to allow adoption of GH. The World Bank is supporting GH financing in Chile, Romania, Türkiye, and Morocco, and is providing TA in Tunisia, Kenya, Brazil, Uzbekistan, Colombia, and Costa Rica.

This DPO is complemented by the World Bank's support to the government's broader energy transition program. At the request of the GoI, the World Bank has been providing significant technical and financing support for (i) scaling up RE and its supporting infrastructure; (ii) reforming the electricity distribution sector; (iii) improving demand side energy efficiency; (iv) decarbonizing the hard-to-abate industrial and transport sectors; and (v) mitigating the impact of energy transition and coal phase down.

The program is aligned with the Country Partnership Framework (CPF) FY18-22 discussed by the World Bank Group (WBG) Board of Executive Directors on September 20, 2018 (Report No. 126667-IN).²³ In particular, the program is linked with two focus areas: (a) promoting resource-efficient growth and (b) enhancing competitiveness and enabling job creation. Through the first focus area, the WBG has committed to supporting the sectors and areas that are critical for facilitating growth and poverty reduction while promoting greater resource efficiency and increase in access to sustainable energy. Through the second focus area, the WBG is aspiring to strengthen key enablers for job creation and to improve competitiveness through better business and policy environments, strengthened workforce skills, and increased firm capabilities. The DPO series is also fully

²³ The CPF for India expired at the end of FY22. The World Bank is currently preparing a Performance and Learning Review, which will seek to extend the CPF through mid-FY24.



aligned with the WBG Green, Resilient and Inclusive Development framework, and the WBG Climate Change Action Plan 2021–2025. The operation is also fully aligned with the WBG Global Crisis Response Framework (GCRF), where all the prior actions support the adoption of “climate smart policies and incentives” under the broad theme of strengthening policies, institutions and investments for rebuilding better (Pillar 4 of the GCRF framework).



ANNEX 7: CHANGES TO THE ORIGINAL MATRIX

Original Pillars, Indicative Triggers, and Indicators	Updated DPF Results Framework	Comments
PILLAR 1: Promoting green hydrogen	Changes to RI1 and RI2	
Results Indicator #1: GH production capacity incentivized (million tonnes) Baseline: [0] [2022] Target: [3] [2026]	Target lowered to 450,000 tons	The targets are adjusted because this is the actual bid that the government has issued. The government's incentives will be provided in FY2025/26 when the winning bidders produce GH.
Results Indicator #2: Domestic electrolyzer manufacturing capacity incentivized (GW) Baseline: [0] [2022] Target: [3] [2026]	Target lowered to 500MW	The earlier estimates were based on the NGHM targets of 5 million tons of GH by 2030, in consultation with the government. But the recently issued government incentive scheme clarified that the incentives will be provided to 450,000 tons of GH production and 500 GW of electrolyser manufacturing each year for FY25/26 to FY29/30.
PILLAR 2: Scaling up renewable energy	One prior action added	
Indicative Trigger: NA	Revised to Prior Action # 6: "To reduce BESS costs, Cabinet has approved a policy providing a financial incentive of up to 40 percent of the capital costs of BESS as viability gap fund for 4000 MWh of BESS by 2030-31".	The newly added PA6 supports incentives for the deployment and use of battery storage. This is a breakthrough policy that will help bring down the costs of battery storage. As soon as RE plus storage reaches cost parity with coal power, coal power plants will no longer be required.
PILLAR 3: Enhancing climate finance for low-carbon energy investments	No Change	N/A